

Final Results — Bruise Segmentation

Injury Analytics Lab

Source: `bruise_colab_final.ipynb + results_final/`

Test set: 185 white-light images / 28 subjects (strict 3-labeler intersection), scored against the **2-of-3 majority vote** on a common 640×640 grid. **YOLO evaluation:** native Ultralytics `.predict()` argmax.

Metric note: Dice is strongly bimodal (a model either localizes the bruise or blanks). *Median* = typical-case quality; *mean* is dragged down by complete misses; *miss %* = the safety axis (images with a bruise where the model predicts *zero* pixels).

1 Accuracy — test results (185 images)

Model	Params (M)	Mean Dice	Median Dice	Miss %
SegFormer-B2 (teacher)	27.35	0.7692	0.8192	0.00
SegFormer-B0 (distilled, $\alpha=0.6$)	3.71	0.7680	0.8167	0.00
SegFormer-B0 (direct)	3.71	0.7663	0.8129	0.54
YOLO26n (distilled, $\alpha=0.4$)	1.63	0.7261	0.8012	2.16
YOLO26n (direct)	1.63	0.7021	0.8061	6.49

The three SegFormers are statistically indistinguishable on Dice (~ 0.77 mean / ~ 0.81 median) but separate sharply on **miss rate**: the teacher and the distilled student never blank. YOLO's *median* is competitive (~ 0.80) but its *mean* is lower because it blanks on 2–6 % of images.

2 External baselines (val-selected best seed — same policy as the 5 models above)

Model	Params (M)	Mean Dice	Median Dice	Miss %
DeepLabV3+ (ResNet-50)	26.68	0.7584	0.8183	2.16
U-Net (ResNet-50)	32.52	0.7570	0.8329	3.78

Both baselines are now selected the **same way as the five core models** — highest *validation* Dice across 3 seeds, then that seed's test result — so the comparison is symmetric (best-of-3 for everyone). They land just below the three SegFormers on mean Dice and blank more often; the SegFormer distilled student remains the best of everything tested and, at 3.71 M params, one of the smallest. (U-Net posts the highest *median* Dice of any model, 0.8329, but its larger mean-vs-median gap and 3.78 % miss rate reflect more complete failures.)

3 Speed / latency (640-tensor-on-GPU $\rightarrow$ mask, per image)

Timed inside a double `cuda.synchronize()`; disk / decode / resize / copy excluded (identical across models, I/O-dominated).

Model	Median (ms)	p95 (ms)	FPS	Params (M)	Peak act. (MB)
SegFormer-B2 (teacher)	33.67	34.95	29.7	27.35	1779
SegFormer-B0 (direct)	16.68	16.94	59.9	3.71	1204
SegFormer-B0 (distilled)	16.64	17.07	60.1	3.71	1204
YOLO26n (direct)	8.18	8.36	122.2	1.63	967
YOLO26n (distilled)	8.22	8.51	121.7	1.63	967

YOLO is $\sim 2\times$ faster and $\sim 7\times$ smaller than B2; the B0 student matches the B2 teacher’s accuracy at $2\times$ the teacher’s speed.

4 Paired contrasts (subject-level cluster bootstrap)

Δ = model A – model B on the same resample. $P(\Delta > 0)$ is one-sided.

Contrast	Δ	95% CI	$P(\Delta > 0)$	Verdict
Distillation (B0-dist – B0-direct)	+0.0017	[-0.009, +0.013]	0.58	n.s.
Student – teacher (B0-dist – B2)	−0.0012	[-0.020, +0.018]	0.44	n.s.
YOLO distilled – direct	+0.0239	[-0.014, +0.059]	0.90	n.s.
SegFormer – YOLO (B0-dist – YOLO-dir)	+0.0659	[+0.011, +0.121]	0.994	significant
Miss %: YOLO-direct – B0-dist	+6.49	[+2.39, +11.63]	1.00	significant

At $n=28$ subjects, Dice differences between healthy models (distillation, student-vs-teacher) are smaller than the resolvable effect — they are null. What *is* resolvable lives in the **miss rate** and in the SegFormer-vs-YOLO gap.

5 Fairness across skin tone (ITA groups)

Per-image Dice averaged over runs, stratified by 5 ITA groups. Kruskal–Wallis omnibus + Mann–Whitney (Bonferroni over 10 pairs). Fairness gap = best – worst group median.

Model	Omnibus p	Sig.?	Dice gap	Worst group	Miss gap
SegFormer-B2 (teacher)	0.011	significant	0.112	Tan (IV)	0.00
SegFormer-B0 (direct)	0.638	n.s.	0.052	Dark (VI)	0.00
SegFormer-B0 (distilled)	0.470	n.s.	0.050	Dark (VI)	0.00
YOLO26n (direct)	0.230	n.s.	0.104	Light (II-III)	0.154
YOLO26n (distilled)	0.151	n.s.	0.090	Light (II-III)	0.051

Only one pairwise comparison survives Bonferroni: SegFormer-B2 Intermediate (III-IV) vs Dark (VI), $p=0.0079$. Everything else is n.s. after correction. YOLO’s worst group is **Light**, not Dark — driven by a size confound (light-skin bruises are smaller, so more misses) — and the intervals are too wide at 9–17 subjects/group to report as a fairness finding. At $n=28$, subgroup gaps are not reliably resolvable.

Bottom line

1. **Ship SegFormer-B0 distilled** — matches the B2 teacher’s accuracy (Δ n.s.) at $2\times$ the speed and $7\times$ fewer params, and never blanks (0/185).
2. **Keep B2 teacher** as the distillation source and a model that also never blanks (0/185).
3. **YOLO** is $2\times$ faster / $7\times$ smaller but blanks on 2–6 % of images $\rightarrow$ risky for injury documentation despite a competitive median. If reported, lead with **miss rate**, not Dice.
4. **Model differences are label-limited:** at $n=28$ subjects, the only resolvable accuracy signals are the miss rate and the SegFormer-vs-YOLO gap; healthy-model Dice differences are null.